Reference: Arakawa's lectures, 1605.00138.

Idea behind slices.

Example

$X = \mathrm{Mat}_{n\times n}(\mathbb{C})$, $G = GL_n(\mathbb{C})$.

$G \curvearrowright X$ by conjugations.

$S = \left\{ \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix} \middle| \lambda_i \neq \lambda_j, i\neq j \right\} \cong \mathbb{C}^n$.

Then the restriction

$$\mathbb{C}[X]^G \longrightarrow \mathbb{C}[S] \cong \mathbb{C}[\lambda_1,\dots,\lambda_n].$$

is injective.

It maps to $\mathbb{C}[\lambda_1,\dots,\lambda_n]^{S_n}$ onto

- Actually it is surjective (as $\mathbb{C}[\lambda_1,\dots,\lambda_n]^{S_n} \cong \mathbb{C}[e_1,\dots,e_n]$).

§ Companion matrices and invariant polynomials.

Set $G = GL_n$, $\mathfrak{g} = \mathfrak{gl}_n$.

Consider

$$\mathcal{C} = \left\{ \underbrace{\begin{pmatrix} 0 & 0 & \cdots & 0 & -a_1 \\ 1 & 0 & \cdots & 0 & \vdots \\ & \ddots & \ddots & 0 & \vdots \\ 0 & & & 1 & -a_n \end{pmatrix}}_{\alpha} \middle| a_i \in \mathbb{C} \right\} \text{ - Companion matrices.}$$

Properties:

- $\det(\lambda - \alpha) = a_1 + a_2\lambda + \dots + a_n\lambda^{n-1} + \lambda^n$.
- $G.\mathcal{C} = \{\text{cyclic operators in } \mathrm{Mat}_{n\times n}\}$.
- $G.\mathcal{C}$ is open dense in $\mathrm{Mat}_{n\times n}(\mathbb{C})$.
- Any G-orbit intersects $\mathcal{C}$ in at most one point.

Theorem

The restriction map

$$R: \mathbb{C}[\mathfrak{g}]^G \longrightarrow \mathbb{C}[\mathcal{C}].$$

Is an isomorphism.

Proof.

$\forall f \in \mathbb{C}[\mathfrak{g}]^G$ if $f|_{\mathcal{C}} = 0 \Rightarrow f(A) = 0\ \forall$ cyclic operator $\Rightarrow$
$\Rightarrow f = 0. \Rightarrow R$ is injective.
$\forall \tilde{f} \in \mathbb{C}[\mathcal{C}]$, define $f(x) = \tilde{f}(c_1(x),\dots,c_n(x))$, where
$c_i(x)$ is defined by $\det(\lambda - x) = \sum_{i=1}^{n-1} c_i(x)\lambda^i + \lambda^n$.
Then clearly $f|_{\mathcal{C}} = \tilde{f}$ as $c_i\left(\begin{pmatrix} 0 & & -a_1 \\ 1 & 0 & \vdots \\ 0 & 1 & -a_n\end{pmatrix}\right) = a_i$. ■

Note that

$\mathcal{C} = f + \mathfrak{a}$, where $f = \begin{pmatrix} & & 0 \\ 1 & & \\ 0 & \ddots\ 1 & 0 \end{pmatrix}$; $\mathfrak{a} = \left\{\begin{pmatrix} 0 & \begin{matrix} * \\ \vdots \\ * \end{matrix}\end{pmatrix}\right\}$.

Fix standard $\mathfrak{n}, \mathfrak{b}$ and pairing by $\mathrm{Tr}(\cdot\ \cdot)$ on $\mathfrak{g}$.

Define $\chi: \mathfrak{n} \longrightarrow \mathbb{C}$
$x \longrightarrow (f, x) = \mathrm{Tr}(f.x)$.

Note: $\chi([\mathfrak{n},\mathfrak{n}]) = 0$.

- $\mathcal{O}_N(\chi) = \{\chi\}$.
- Let $\mu: \mathfrak{g}^* \longrightarrow \mathfrak{n}^*$ be restriction map.
Then $\mu^{-1}(\chi) = \chi + \mathfrak{n}^{\perp} \cong f + \mathfrak{b}$ is N-invariant set.

Theorem (Kostant)

The action $N \times \mathcal{C} \xrightarrow{\varphi} f + \mathfrak{b}$
$(g, x) \longmapsto g.x.g^{-1}$

Is an N-equivariant isomorphism (of affine varieties)

Proof.

$[\mathfrak{n}, f+\mathfrak{b}] \subset \mathfrak{b} \Rightarrow N.(f+\mathfrak{b}) \subset f + \mathfrak{b}$.

- Any matrix in $f+\mathfrak{b}$ is cyclic hence φ is surjective.
- Two companion matrices are conjugated iff they are equal.

Stabilizer of a companion matrix in N is trivial:

$\forall \xi \in \mathfrak{n}$ decompose ξ w/ $\mathfrak{sl}_2$ triple (e,h,f) weights.

Then $\xi = \sum_{i>0} \xi_i$ let i_0 be minimal s.t. $\xi_{i_0} \neq 0$.

Then $[f+\mathfrak{a}, \xi] = \underbrace{[f, \xi_{i_0}]}_{\neq 0} + \text{higher weights}$.

Hence $Z_{\mathfrak{n}}(f+\mathfrak{a}) = \{0\} \Rightarrow Z_N(f+\mathfrak{a})$ is finite $\Rightarrow$ trivial.

Then if $\varphi((g,x)) = \varphi((\tilde{g},\tilde{x})) \Rightarrow$ 1) $x = \tilde{x}$
2) $g.\tilde{g}^{-1} \in Z_N(x) \Rightarrow g = \tilde{g}$.

Hence φ is injective.

We conclude that φ is bijection.

- Then φ is birational (some alg. geometry)
- Since $f+\mathfrak{b}$ is normal, φ^{-1} is regular (Zarissky main theorem) ■

Corollary.

The restriction maps

$$\mathbb{C}[\mathfrak{g}]^G \xrightarrow[\varphi_1]{} \mathbb{C}[f+\mathfrak{b}]^N \xrightarrow[\varphi_2]{} \mathbb{C}[\mathcal{C}]$$

Are isomorphisms.

Proof. $\varphi = \varphi_2 \circ \varphi_1$ is an isomorphism $\Rightarrow \varphi_2$ is surjective, φ_1 is inj.
$\forall f \in \mathbb{C}[f+\mathfrak{b}]^N \Rightarrow f|_{\mathcal{C}} = 0 \Rightarrow f|_{f+\mathfrak{b}} = 0$ as $f + \mathfrak{b} = N.\mathcal{C} \Rightarrow \varphi_2$ is inj.
$\forall f \in \mathbb{C}[f+\mathfrak{b}]^N \Rightarrow f = \varphi_1(\varphi^{-1}(\varphi_2(f)))$ ■

§ Transversality.

Lemma The space $f+\mathfrak{b}$ intersect $\mathcal{O}_G(f)$ at f transversely.

Proof:

Due to $\mathfrak{sl}_2$-decomposition.

$T_f(f+\mathfrak{b}) = \mathfrak{b}$, $T_f(\mathcal{O}_G) = [f,\mathfrak{g}] \supset \mathfrak{n}_- \Rightarrow T_f(f+\mathfrak{b}) + T_f(\mathcal{O}_G) = \mathfrak{g}$ ■

Def.

Let (e,h,f) be an $\mathfrak{sl}_2$-tripple corresp. to f.

This defines embedding $\tilde{\gamma}: SL_2(\mathbb{C}) \hookrightarrow G$.

Then $\gamma: \mathbb{C}^\times \longrightarrow G$ is defined as $\tilde{\gamma}|_{\mathbb{C}^\times = \{\left(\begin{smallmatrix} t & 0 \\ 0 & t^{-1}\end{smallmatrix}\right) | t\in\mathbb{C}^\times\} \subset SL_2}$.

Define $\rho: \mathbb{C}^\times \times \mathfrak{g} \longrightarrow \mathfrak{g}$
$(t, x) \longmapsto t^2 \mathrm{Ad}_{\gamma(t)}(x)$.

Exercise ρ contracts $f + \mathfrak{b}$ to f. Namely
$\forall x \in f+\mathfrak{b}$, $\lim_{t\to 0}(\rho(t,x)) = f$.

Prop. The affine space $f+\mathfrak{b}$ intersects adjoint G-orbits transversely at any point of $f+\mathfrak{b}$.

Proof:

This is true at $f \Rightarrow$ holds in a neighborhood (usual top.) of f.
But $f+\mathfrak{b}$ is cotracted to f by torus action and this action preserves transversality. ■

§ $S = \mathcal{C}$ as a reduction.

Recall: $\mathfrak{g}^*$ is a Poisson variety w/ KKS bracket
$\{x,y\} = [x,y]\ \forall x,y \in \mathfrak{g} = (\mathfrak{g}^*)^*$.

Lemma:

- The coadjoint action $G \curvearrowright \mathfrak{g}^*$ is hamiltonian w/ moment map $\mu = \mathrm{Id}: \mathfrak{g}^* \longrightarrow \mathfrak{g}^*$.
- Let $G \curvearrowright X$ be a hamiltonian action on X w/ $\mu^G: X \to \mathfrak{g}^*$. Let $H \leq G$, then restriction $H \curvearrowright X$ is also hamiltonian w/ $\mu^H = r^{\mathfrak{g}}_{\mathfrak{h}} \circ \mu^G$, $r^{\mathfrak{g}}_{\mathfrak{h}}: \mathfrak{g}^* \longrightarrow \mathfrak{h}^*$ - restriction, $\mathfrak{h} = \mathrm{Lie}(H)$.

■ (Exercise)

Consider adjoint action of $N \curvearrowright \mathfrak{g}^*$.

Then it is hamiltonian w/ moment map given by restriction $r = \mu: \mathfrak{g}^* \longrightarrow \mathfrak{n}^*$.

As we've seen:

$\mathcal{O}_N(\chi) = \{\chi\}$

Then $\mathfrak{g}/\!\!/_{\mathcal{O}_N(\chi)} N = \mu^{-1}(\mathcal{O}_N(\chi))/N = \mu^{-1}(\chi)/N = (f+\mathfrak{b})/N =$
$= (S \times N)/N \cong S$.

Cor. S is a Poisson variety.

Q: What is the bracket?

Recall: We have

$\mathbb{C}[X/\!\!/_0 G] = \mathbb{C}[\mu^{-1}(0)]^G$, Poisson bracket induced from by restricting from X.

Hence $\mathbb{C}[\mathfrak{g}/\!\!/_\chi N] = \mathbb{C}[\chi + \mathfrak{n}^\perp]^N = \mathbb{C}[f+\mathfrak{b}]^N \cong \mathbb{C}[\mathfrak{g}]^G$.

But $\forall f_1, f_2 \in \mathbb{C}[\mathfrak{g}]^G$, $\{f_1, f_2\} = 0 \Rightarrow$ We get 0 bracket.
Why?

- Move towards generalization.

§ Kostant slices.

Prop. $\mathcal{C} \cong f + Z_{\mathfrak{g}}(e) \subset f + \mathfrak{b}$.

Proof.

$\forall x \in f + Z_{\mathfrak{g}}(e) \subset f + \mathfrak{b}$ by Kostant's theorem exists a unique pair $(g, a) \in N \times \mathcal{C}$: $g.a.g^{-1} = x$.

This gives a map

$$\begin{tikzcd} f + Z_{\mathfrak{g}}(e) \arrow[r, hook, "z"] \arrow[rd, "\Psi"'] & f+\mathfrak{b} \cong N\times\mathcal{C} \arrow[d, "\pi_2"] \\ & \mathcal{C} \end{tikzcd}$$

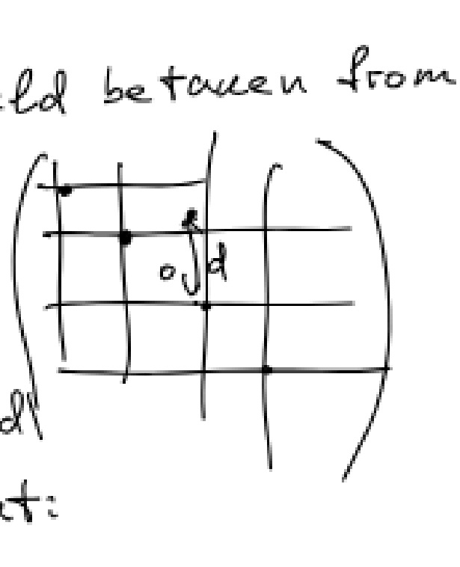
$x \in f + Z_{\mathfrak{g}}(e) \Rightarrow x \sim \begin{pmatrix} b_0 & b_1 & \cdots & b_{n-1} \\ (n-1) & b_0 & \ddots & \vdots \\ & 2(n-2) & \ddots & b_1 \\ 0 & & (n-1)\cdot 1 & b_0 \end{pmatrix}$.

We study $\det(\lambda - x)$.

Claim: The coeff of λ^{n-k} has the form (Here $k \geq 1$)
$\underset{\neq 0}{c_k} b_{k-1} + p_k(b_1,\dots,b_{k-2})$.

Proof.

To get λ^{n-k} at least $n-k$ terms should be taken from the diagonal. We have $\sum_{i=1}^n \sigma(i) = \frac{n(n+1)}{2}$.



Hence if b_m appears $\Rightarrow$ for at least d indices i: $\sigma(i) = i+1$.

Hence maximal possible m is k and for such term we have σ such that:

1) σ has $n-k$ fixed points.
2) $\exists\ i_1 < \dots < i_{k-1}$: $\sigma(i_j) = i_j + 1$.

This implies that σ has the form
$(i\ i{+}1 \dots i{+}k{+}1)$.

All such go with coeff. of the same sign by b_{k-1}. ■

Therefore the map Ψ is described by
$a_{n-k} = \underset{\neq 0}{c_k} b_{k-1} + p_k(b_0,\dots,b_{k-2}) \Rightarrow$ is an isomorphism. ■

Def. $f + Z_{\mathfrak{g}}(e)$ is called Kostant slice.

Remarks.

- Kostant slice is also transverse to orbits.

§ Slodowy slices.

Def. Let $e \in \mathfrak{g}$ be a nilpotent element.
Let (e,h,f) be a corresp $\mathfrak{sl}_2$-triple.
$S_f := f + Z_{\mathfrak{g}}(e)$.

Proposition.

(i) S_f is transverse to orbits $\mathcal{O}_x$, namely, if $x \in S_f$, $T_x\mathcal{O}_x + T_x S_f = T_x\mathfrak{g} \cong \mathfrak{g}$.

(ii) $S_f \cap \mathcal{O}_f = \{f\}$.

Proof.

(i) First for $x = f$.
$T_f\mathcal{O}_f = \mathrm{Im}(\mathrm{ad}_f)$, $T_f S_f = Z_{\mathfrak{g}}(e) = \mathrm{Ker}(\mathrm{ad}_e)$.
For any f.d. $\mathfrak{sl}_2$-rep: $W = \mathrm{Im}(\rho(f)) \oplus \mathrm{Ker}(\rho(e))$.
In general, $x = f + s$, $s \in \mathrm{Ker}(\mathrm{ad}(e))$.
$T_x\mathcal{O}_x = \mathrm{Im}(\mathrm{ad}_x)$, $T_x(S_f) = \mathrm{Ker}(\mathrm{ad}_e)$ are transverse in some nbhd of $x \in \mathfrak{g}$.

$\mathfrak{sl}_2$-triple induces a homomorphism $\gamma: SL_2(\mathbb{C}) \longrightarrow G$.
We have adjoint action of $h_t = \begin{pmatrix} t & 0 \\ 0 & t^{-1}\end{pmatrix}$
$\mathrm{Ad}_{\gamma(h_t)}(f) = t^{-2}.f$.
Define $\mathbb{C}^\times$ action on S_f by
$t: f + s \longmapsto t^2.\mathrm{Ad}_{\gamma(t^{-1})}(f+s) = f + t^{-2}\mathrm{Ad}_{\gamma(t^{-1})}(s)$.
$s \in \mathrm{Ker}(\mathrm{ad}(f)) \Rightarrow h_t$ weights are $\leq 0 \Rightarrow t$-weights of s are $> 0 \Rightarrow \lim_{t\to 0}(t.(f+s)) = f\ \forall f + s \in S_e$.

Therefore since the action preserves transversality, in a nbhd intersection is transverse and action contracts S_f to $f \Rightarrow$ ■

(ii) Fact: $\mathcal{O}_f$ is $\mathbb{C}^\times$ stable

Note that $\mathcal{O}_f$ is $\mathbb{C}^\times$-invariant $\Rightarrow t^{-2}\,\mathrm{Ad}_{\gamma(t^{-1})}(\mathcal{O}_f) \subset \mathcal{O}_f$.
Moreover S_f is $\mathbb{C}^\times$-invariant.
Then $\mathcal{O}_f \cap S_f$ is $\mathbb{C}^\times$ invariant.
Then if $x \in \mathcal{O}_f \cap S_f \setminus \{f\}$, $\lim_{t\to 0}(t.x) = f \Rightarrow$
$\mathcal{O}_f \cap S_f$ intersects arbitrarily small nbhds of f / there is a curve on $\mathcal{O}_f \cap S_f$ through f.
But $\mathcal{O}_f$ and S_f are strictly transverse at f. ■

Remark: Away from f intersection in general is not strictly transverse.
Dimension of $\mathcal{O}_G(f)$ is complementary to $\dim(S_f)$.
But this is not the case for $\mathcal{O}_G(x)$, $x \in S_f$ in general.

Example $G = SL_2$, $\mathfrak{g} = \mathfrak{sl}_2$

$S_f = \{f + x.e \mid x \in \mathbb{C}\}$.

Generic orbits are defined by:

$\mathcal{O}_\Delta = \left\{\begin{pmatrix} a & b \\ c & -a\end{pmatrix} \middle| a^2 + bc = \Delta\right\}$

Intersections: $c = 1, a = 0 \Rightarrow x = b = \Delta$ - unique int.
On $\mathcal{O} = \{0\} \Rightarrow$ no intersections.

$T_{\left(\begin{smallmatrix} 0 & x \\ 1 & 0\end{smallmatrix}\right)}(\mathcal{O}) = \mathrm{Ker}(db + x\,dc)$

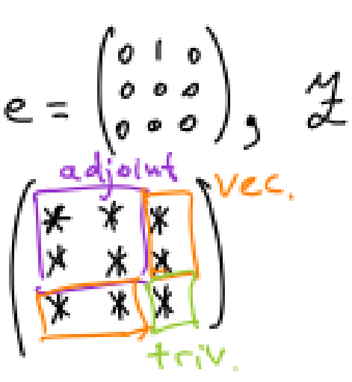
$T_{\left(\begin{smallmatrix} 0 & x \\ 1 & 0\end{smallmatrix}\right)}(S_f) = \langle e \rangle = \langle \frac{\partial}{\partial b} \rangle$.

$\langle db + x\,dc, \frac{\partial}{\partial b}\rangle \equiv 1 \Rightarrow$ Always transverse.

Example: $G = SL_3(\mathbb{C})$, $\mathfrak{g} = \mathfrak{sl}_3(\mathbb{C})$, subregular orbit.

$f = \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0\end{pmatrix}$, $e = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0\end{pmatrix}$, $Z_{\mathfrak{g}}(e) = \left\langle e, \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 2\end{pmatrix}, \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0\end{pmatrix}, \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 1 & 0\end{pmatrix}\right\rangle$

$\mathfrak{sl}_2$ decomp:



Hence $S_f = \left\{ X = \begin{pmatrix} -x_0 & x_3 & x_1 \\ 1 & -x_0 & 0 \\ 0 & x_2 & 2x_0\end{pmatrix} \middle| x_i \in \mathbb{C}\right\}$.

$\mathrm{Tr}(X^2) = 6x_0^2 + x_3 + x_1x_2$; $\det(X) = 2x_0^3 + x_1x_2 - 2x_0x_3$.

§ Poisson structure on S_f via reduction.

- Decompose $\mathfrak{g} = \bigoplus_{i\in\mathbb{Z}} \mathfrak{g}_i$ - h-weight decomposition.
- Define $\omega_s: \Lambda^2\mathfrak{g}_1 \longrightarrow \mathbb{C}$
$x \wedge y \longmapsto \chi([x,y]) = \langle f, [x,y]\rangle$

Prop. ω_s is a symplectic 2-form on $\mathfrak{g}_1$.

Proof. Non-degeneracy:
$x \in \mathfrak{g}_1^{\angle} \Leftrightarrow [f,x] \in \mathfrak{g}_1^{\perp}$.
But $\mathrm{ad}_f: \mathfrak{g}_1 \xrightarrow{\sim} \mathfrak{g}_{-1}$ and $\langle\cdot,\cdot\rangle|_{\mathfrak{g}_1\otimes\mathfrak{g}_{-1}}$ is non-deg. $\Rightarrow x = 0$
■.

- Terminology/Notation:

Let V be a vector space, ω a symplectic form on V, then for a subspace $W \subset V$,

- $W^{\angle} := \{u \in V \mid \omega(u,w) = 0\ \forall w \in W\}$.

A subspace $W \subset V$ is

- Isotropic if $W \subseteq W^{\angle}$.
- Coisotropic if $W^{\angle} \subseteq W$.
- Lagrangian if $W^{\angle} = W$ (isotropic and coisotropic).

Fix ℓ - any lagrangian subspace of $\mathfrak{g}_1$.

Remark If f-principal nilpotent, $\mathfrak{g}_1 = \{0\}$.

Set $\mathfrak{m} = \ell \oplus \bigoplus_{i\geq 2}^{\infty} \mathfrak{g}_i$. Set $M \subset G$: $\mathrm{Lie}(M) = \mathfrak{m}$, M is connected.

- $\mathfrak{m}$ is a nilpotent Lie algebra.